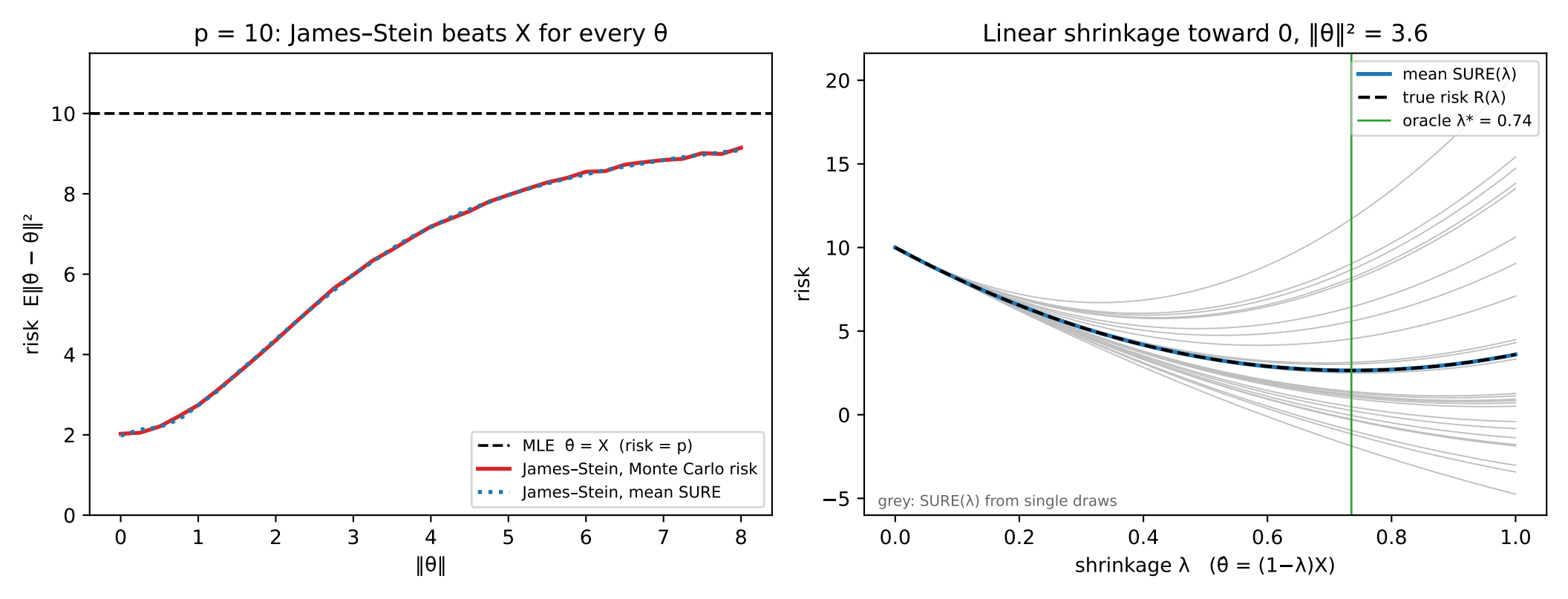
p = 10: James–Stein beats X for every θ
risk E‖θ − θ‖²
‖θ‖
MLE θ = X (risk = p)
James–Stein, Monte Carlo risk
James–Stein, mean SURE
Linear shrinkage toward 0, ‖θ‖² = 3.6
risk
shrinkage λ (θ = (1−λ)X)
mean SURE(λ)
true risk R(λ)
oracle λ* = 0.74
grey: SURE(λ) from single draws